CSS @ GLANCE

Structure or Syntax of CSS

- ✓ The syntax of CSS is slightly different from that of an HTML.
- ✓ CSS uses **(curly braces { }), (colons :)** and **(semicolon ;)**.

Syntax:

```
selector
{
        property : value;
        property : value;
        |                       |
        property : value;
}
```

Importance of semicolon

```
p{text-align:center background-color:tomato ;color:cyan}
```

• The code has a syntax error (missing semicolon after text-align:center).
• Because of the error, the browser ignores the entire line until the next semicolon.
• Only the last valid property works:
Color:cyan;
• Output: The <p> text becomes cyan color only , no background-color and align applied

Types of CSS

Type of CSS	Description	Where It Is Written	Syntax (Example)
1. Inline CSS	Used to style a **single HTML element** directly. It has the **highest priority**.	Inside the HTML tag using the style attribute.	Heading
2. Internal (Embedded) CSS	Used to style elements **within the same HTML page**.	Inside the <style> tag in the <head> section of the HTML document.	<head><style> p { color: red; font-size: 18px; } </style></head>
3. External CSS	Used to apply styles to **multiple web pages** using a separate .css file.	In an external stylesheet linked using the <link> tag in the <head> section.	<head><link rel="stylesheet" href="style.css"></head>*(style.css file contains CSS rules)*

Various CSS Selectors

Selector Type	Description	Syntax	Example
Element Selector	Selects all elements of a specific type/tag. Lowest specificity (except universal).	**element**	p { color: black; }
ID Selector	Selects a single element with a specific id attribute. Very high specificity.	**#id**	#header { color: red; }
Class Selector	Selects elements with a specific class. Medium specificity.	**.class**	.menu { font-size: 16px; }
Attribute Selector	Selects elements based on an attribute or attribute value.	**[attr], [attr=value]**	input[type="text"] { border: 1px solid #ccc; }
Universal Selector	Selects all elements. Lowest specificity.	*****	* { margin: 0; padding: 0; }
Descendant Selector	Selects elements that are descendants (any level) of a specified ancestor.	**ancestor descendant**	div p { color: blue; }
Child Selector	Selects elements that are direct children of a specified parent.	**parent > child**	ul > li { list-style: none; }
Grouping Selector	Groups multiple selectors and applies the same styles.	**selector1, selector2**	h1, h2, h3 { font-family: Arial; }

CSS Specificity (Priority Order)

When multiple CSS rules target the same element, the browser decides which one to apply based on **specificity** (priority):

Priority Level	Selector Type	Example
1 (Highest)	Inline Styles	Heading
2	IDs	#title { color: red; }
3	Classes, pseudo-classes, attribute selectors	.title { color: green; }, :hover, [type='text']
4 (Lowest)	Elements and pseudo-elements	h1 { color: blue; }, ::before

Text Properties

These properties are used to **style, format, and control the appearance** of text on a webpage, including alignment, spacing, transformation, decoration, font type, and size.

Text Properties

Property	Description	Syntax	Possible Values
color	Sets the color of the text	color: color;	Named colors (e.g., red), HEX (#ff0000), RGB (rgb(255,0,0)), RGBA
text-align	Aligns text inside an element	text-align: value;	left, right, center, justify, start, end
text-indent	Adds indentation to the first line	text-indent: length;	Any length (px, em, %)
text-transform	Controls text capitalization	text-transform: value;	none, capitalize, uppercase, lowercase
text-decoration	Adds decoration like underline or line-through	text-decoration: value;	none, underline, overline, line-through, underline overline
letter-spacing	Controls space between letters	letter-spacing: length;	Normal (normal), or custom spacing (px, em). Negative values are allowed.
word-spacing	Controls space between words	word-spacing: length;	Normal (normal), or custom spacing (px, em). Negative values are allowed.
line-height	Sets space between lines	line-height: value;	normal, number (e.g., 1.5), length (px, em), %. Negative values are **not** allowed.
text-shadow	Adds shadow to text	text-shadow: h-shadow v-shadow blur color;	none or values like 2px 2px 5px gray (can add multiple shadows separated by commas)

Font Properties

Property	Description	Syntax	Example	Possible Values
font-family	Specifies the font type	font-family: "FontName", fallback;	font-family: "Arial", sans-serif;	Any font name: "Arial", "Times New Roman", "Courier New", generic families: serif, sans-serif, monospace, cursive, fantasy
font-size	Sets the font size	font-size: size;	font-size: 20px;	Length units: px, em, rem, %; Keywords: xx-small, x-small, small, medium, large, x-large, xx-large, smaller, larger
font-style	Defines style of text	font-style: value;	font-style: italic;	normal, italic
font-weight	Sets font thickness	font-weight: value;	font-weight: bold;	normal, bold, bolder, lighter, 100, 200, 300, 400, 500, 600, 700, 800, 900
font-variant	Displays text in small-caps	font-variant: value;	font-variant: small-caps;	normal, small-caps
font (shorthand)	Sets all font properties in one line	font: [style] [variant] [weight] [size]/[line-height] [family];	font: italic small-caps bold 16px/1.5 "Times New Roman", serif;	Combines all above properties in one; values are the same as individual properties: <style> <variant> <weight> <size>/<line-height> <family>

Google Fonts

How to Embed Google Fonts

Method 1: Using <link> tag (Recommended)

Method 2: Using @import in CSS

CSS Borders

A **border** is a line that wraps around an HTML element's **content and padding**. It visually separates elements or highlights them on a web page.

Property	Description	Possible Values	Syntax
border-style	Defines the style of the border (Must to Define)	none, solid, dashed, dotted, double **Note:** Multiple values allow to get different border side wise	border-style: solid border-style: solid dashed double dotted
border-width	Sets the thickness of the border	length (px, em, rem)	border-width: 10px
border-color	Sets the color of the border	Named colors (red), hex (#FF0000), RGB (rgb(255,0,0))	border-color: tomato
border (shorthand)	Combines style, width, color in one line	border: 2px solid red;	border: 2px solid red;
border-top	Sets border properties for the top side	border-top: 3px dashed blue;	border-top: 3px dashed blue;
border-right	Sets border properties for the right side	border-right: 2px solid green;	border-right: 2px solid green;
border-bottom	Sets border properties for the bottom side	border-bottom: 4px dotted orange;	border-bottom: 4px dotted orange;
border-left	Sets border properties for the left side	border-left: 5px double purple;	border-left: 5px double purple;

Box Properties

Property	Description	Possible Values	Syntax / Examples
margin	Creates **space outside** the element (outside border). **Used separate elements** on the page.	• auto → browser calculates margin (useful for centering) • length → px, em, %, negative values allowed	**margin: 10px** **Shorthand examples:** • 4 values → margin: 25px 50px 75px 100px; → top=25px, right=50px, bottom=75px, left=100px • 3 values → margin: 25px 50px 75px; → top=25px, right/left=50px, bottom=75px • 2 values → margin: 25px 50px; → top/bottom=25px, right/left=50px • 1 value → margin: 25px; → all sides=25px
padding	Creates **space inside** the element (between content & border).	length → px, em, %	**padding:10px** **Shorthand examples:** • 4 values → padding: 25px 50px 75px 100px; → top=25px, right=50px, bottom=75px, left=100px • 3 values → padding: 25px 50px 75px; → top=25px, right/left=50px, bottom=75px • 2 values → padding: 25px 50px; → top/bottom=25px, right/left=50px • 1 value → padding: 25px; → all sides=25px
width	Specifies the **width** of an element's content area.	auto, length (px, em, %),	width : 50px
height	Specifies the **height** of an element's content area.	auto, length (px, em, %),	height:50px
box-sizing	Defines **how width and height are calculated** — whether padding and borders are included. Used to **control layout behavior** when adding padding/borders.	• content-box (default): width/height exclude padding & border. • border-box: width/height include padding & border.	**box-sizing : content-box** (height and width increase by increasing padding and border px) **box-sizing : border-box** (height and width will remain same)
box-shadow	Adds **shadow effects** around an element's frame. To **add depth**, **hover effects**, or **visual highlights**.	box-shadow: h-offset v-offset blur spread color;	**Syntax:** box-shadow: h-offset v-offset blur spread color; **Example:** box-shadow: 2px 2px 5px gray;

CSS background properties

Property	Description	Key Values / Notes	Syntax
background-color	Sets the background color of an element	Any valid color (red, #ffc0cb, rgb(255,0,0))	background-color: color;
background-image	Sets an image as the background	Use image URL;	background-image: url("image.jpg");
background-repeat	Controls how background image repeats	repeat, repeat-x, repeat-y, no-repeat default repeats if not specified	background-repeat: repeat;
background-position	Sets the starting position of a background image	top, bottom, left, right, center, x% y%	background-position: top;
background-size	Specifies the size of the background image	auto, cover, contain, width height	background-size: auto;
background-attachment	Sets whether background scrolls with content	scroll, fixed, local	background-attachment: scroll;
background-clip	Determines how far the background extends	border-box, padding-box, content-box	background-clip: border-box;
background (shorthand)	Combines color, image, repeat, position, size, attachment	Example: background: pink url("scenary.jfif") no-repeat top right cover fixed;	background: [color] [image] [repeat] [position] [size] [attachment];

Pseudo classes

Pseudo-class	What It Does	Syntax
:hover	Changes color when mouse hovers	Selector:hover{prop:value} p:hover{color:blue;}
:active	Changes color when clicked	a:active{prop:value}
:focus	Highlights input field when clicked	input:focus{prop:value}
:link / :visited	Colors for unvisited/visited links	a:link{prop:value}
:nth-child(n)	Targets the nth list item	li:nth-child(2){prop:value}
:not(.special)	Styles all list items except .special	p:not(.classname){prop:value}

Pseudo Elements

Pseudo-element	Description	Syntax	Example	Result / Effect
::first-line	Styles the **first line** of a block-level element.	p::first-line { property: value; }	p::first-line { color: red; font-weight: bold; }	First line of the paragraph appears red and bold.
::first-letter	Styles the **first letter** of a block-level element.	p::first-letter { property: value; }	p::first-letter { font-size: 50px; color: red; }	First letter of the paragraph becomes large and red.
::before	Inserts content **before** an element's actual content.	selector::before { content: "..."; }	p::before { content: "Note: "; color: red; }	Adds "Note:" before each paragraph.
::after	Inserts content **after** an element's actual content.	selector::after { content: "..."; }	p::after { content: "✔"; color: green; }	Adds a green checkmark after the paragraph.
::marker	Styles the **marker (bullet or number)** of list items.	li::marker { property: value; }	li::marker { color: red; font-size: 20px; }	List bullets or numbers appear red and large.
::selection	Styles the **highlighted text** selected by the user.	::selection { property: value; }	::selection { background: blueviolet; color: white; }	Selected text appears white on a blue-violet background.

Display Property

Display Type	Description	Behavior / Use Case	Visual Behavior	Syntax
inline	Displays elements **in a line**, without starting on a new line.	Does **not accept width/height**. Common for , <a>, .	Elements sit **side by side** in a single line.	display: inline
block	Displays element as a **block**, starting on a new line.	Takes **full width** available and allows **width/height** to be set.	Each element appears **on a new line**.	display: block
inline-block	Combines features of **inline** and **block**.	Appears **inline**, but allows **width and height**.	Boxes are **side by side**, but **size-controllable**.	display: inline-block
none	**Hides** the element completely (removed from layout).	Element takes **no space** on the page.	The element is **invisible** and **does not occupy space**.	display: none
flex	Displays element as a **flex container**.	Allows flexible alignment and distribution of child elements.	Items are **arranged in a row** (or column) with flexible spacing.	display: flex
grid	Displays element as a **grid container**.	Divides layout into **rows and columns** for advanced control.	Elements are placed in a **grid layout** (rows and columns).	display:grid

Flexbox

Property	Description	Possible Values	Syntax / Example
display: flex	Defines a flex container to arrange items flexibly.	flex, inline-flex	div { display: flex; }
flex-direction	Defines the direction of flex items.	row (default), row-reverse, column, column-reverse	flex-direction: row;
justify-content	Aligns items horizontally (along the main axis).	flex-start, flex-end, center, space-between, space-around, space-evenly	justify-content: space-between;
align-items	Aligns items vertically (along the cross axis).	stretch (default), flex-start, flex-end, center	align-items: center;
flex-wrap	Determines whether flex items wrap onto multiple lines.	nowrap (default), wrap, wrap-reverse	flex-wrap: wrap;
gap	Defines the space between flex items.	Any CSS length unit (px, em, %)	gap: 15px;
order	Specifies the display order of flex items.	Integer values (0 default, can be positive or negative)	order: 2;
flex *(shorthand)*	Sets how an item grows, shrinks, and defines its base size.	flex: grow shrink basis; e.g., flex: 1 0 100px;	flex: 1; or flex: 1 0 200px;

Grid Layout Properties

Property	Description	Possible Values	Default Value	Syntax / Example
grid-template-columns	Defines the number and width of columns in a grid layout.	Fixed units (px, em, %)	none	grid-template-columns: 200px 1fr 2fr;
grid-template-rows	Defines the number and height of rows in a grid layout.	Fixed units (px, em, %),	none	grid-template-rows: 100px auto 100px;
gap (grid-gap)	Defines the space between rows and columns.	Any CSS length (px, em, %)	0	gap: 10px; or gap: 20px 40px;
grid-column	Specifies the horizontal position of an item within the grid columns (start / end lines).	<start-line> / <end-line>	auto	grid-column: 1 / 3; (spans across 2 columns)
grid-row	Specifies the vertical position of an item within the grid rows (start / end lines).	<start-line> / <end-line>	auto	grid-row: 1 / 2; (spans across 1 row)

CSS Positioning Properties

Property	Description	Possible Values	Default Value	Syntax / Example
position	Defines how an element is positioned in the document.	static, relative, absolute, fixed, sticky	static	position: absolute;
top	Distance between element and top edge of container.	Any CSS length (px, %, auto)	auto	top: 20px;
right	Distance between element and right edge of container.	Any CSS length (px, %, auto)	auto	right: 10px;
bottom	Distance between element and bottom edge of container.	Any CSS length (px, %, auto)	auto	bottom: 15px;
left	Distance between element and left edge of container.	Any CSS length (px, %, auto)	auto	left: 30px;
z-index	Controls stack order of overlapping elements.	Integer (auto, positive or negative values)	auto	z-index: 2; (Higher = on top)

- ✓ **static: Default, element follows normal document flow.**
- ✓ **relative: Moved relative to its normal position. (Shift top/bottom.. from it's aligned position)**
- ✓ **absolute: Positioned relative to the nearest positioned ancestor**
- ✓ **fixed: Stays fixed in place even when scrolling.**
- ✓ **z-index: Controls which element appears on top of others.**